Rishabh Sahu

Address: Lig 139 Harshwardhan Nagar, Bhopal, Madhya Pradesh, India
Email: rishabhsahu617@outlook.com • **Phone:** (+91) 7898994078
LinkedIn: rishabh-sahu325 • **GitHub:** RishabhSahuIIIT • **GitLab:** rishabhsahu325 •
Portfolio: rishabhsahuiiit.github.io

Education

International Institute of Information Technology Hyderabad — July 2024 – Present
MTech. Computer Science — CGPA: 7.73/10

The LNM Institute of Information Technology Jaipur — July 2019 – June 2023
BTech. Computer Science — CGPA: 7.43/10

G.V.N. The Global School Bhopal — April 2017 – March 2019
High School — 87.2%

Campion School Bhopal — April 2005 – March 2017
Primary and Middle School — CGPA: 10/10

Internships

Web and Mobile app development Intern at C.R.I.S. Secunderabad — 1 June 2025 – 30 July 2025

Keywords: Git, Flutter, Dart, Javascript, Nodejs, MySQL, POSTMAN — **Project repository**

- Worked on a prototype mobile application for Divyangjan card application website
- Implemented backend API routes
- Contributed in database design and testing
- Added frontend pages for Railway division wise reports

Projects

JSON API server — *Backend Development*

Keywords: Flask, Docker, Python, API design, Postgresql — **Project repository**

- Handles external http requests to fetch data from Postgresql Database
- Perform create and view operations for teacher and course data in the database
- Retrieve responses to http requests in JSON format
- Utilized Docker and BASH to reduce setup time

Wine Classifier based on Decision Tree — *Machine Learning*

Keywords: Python, Pyspark, Pandas, Decision Tree classification — **Project repository**

- Generated ML classifier model to classify Wine from characteristics
- Classifier trained on sklearn's wine dataset
- Evaluated trained model through pyspark's model evaluator
- Achieved reasonably good accuracy of 85% on test metrics

N-gram Language Model for Next-Word Prediction — *Machine Learning*

Keywords: Python, Character N-gram, Curses (TUI) — **Project repository**

- Developed a Character-level N-gram model to suggest word completions in real-time as the user types.
- Implemented Maximum Likelihood Estimation (MLE) to calculate transition probabilities between character sequences.
- Built a low-latency Terminal User Interface (TUI) using `curses` that tracks keystroke efficiency metrics live.

Secure many to one communication system with Digital Signatures — *Cryptography, Networking*

Keywords: Security, Python, Cryptography, Network Security, Socket Programming — **Project repository**

- Engineered a secure communication protocol ensuring confidentiality using AES-256 and ElGamal encryption
- Implemented mutual authentication and integrity verification mechanisms using ElGamal Digital Signatures
- Mitigated replay attacks via timestamp validation and enforced access control through temporary blocking
- Orchestrated secure session management with dynamic group key generation for encrypted broadcast messaging

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Multithreaded HTTP Server — *Concurrency, Networking*

Keywords: Systems Programming, Rust, Multithreading, TCP, HTTP, Thread Pool, Concurrency — **Project repository**

- Built a multithreaded HTTP/1.1 server using only Rust's standard library, following the official Rust book's capstone project to pick up the language
- Implemented a fixed-size thread pool that dispatches incoming requests across worker threads
- Handled GET request routing with `200 OK`, simulated latency `/sleep`, and `404 NOT FOUND` responses

- Gained hands-on experience with Rust's ownership model and concurrency primitives (Arc, Mutex, channels) for safe parallel execution

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Shell *Systems Programming*

Keywords: C++, Linux API, Process handling, System calls **Project repository**

- Implemented external program execution through fork and exec
- Developed shell built-ins including `cd`, utilizing OS function calls for directory changes rather than internal state management
- Integrated the readline library for tab-autocompletion, applied specifically to built-in commands
- Orchestrated inter-process communication via pipes
- Manipulated file descriptors to enable output redirection

HTML Autocomplete using Lexer *Compilers, Data Structures*

Keywords: GNU Flex/Lex, C, Lexical Analysis, Trie, Compiler Design **Project repository**

- Implemented an HTML lexer using GNU Flex with custom grammar rules to tokenise HTML documents
- Built a Trie-based string matching engine in C to provide autocomplete suggestions from partial HTML tags and attributes
- Extended token coverage beyond standard HTML spec to support tags like `img`, `header`, and `srcset`

Network Intrusion Detection and Prevention System *Network Security, Concurrency*

Keywords: Security, Systems Programming, Python, Scapy, Network Security, iptables, Multithreading **Project repository**

- Performed real-time TCP packet capture and analysis using Scapy's low-level network sniffing capabilities
- Implemented signature-based detection for port scanning (sliding time-window) and OS fingerprinting (TCP flag pattern matching)
- Automated threat response by dynamically blocking attacker IPs via iptables (Linux) and Windows Firewall (Windows)
- Decoupled packet capture from the management CLI using multi-threading for continuous monitoring without UI blocking

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Agentic AI Research Assistant *Agents, Natural Language Processing*

Keywords: Artificial Intelligence, Python, Moya framework, Ollama, Llama 3.1/3.2, PyMuPDF, Multi-agent systems **Project repository**

- Built a multi-agent pipeline on the Moya framework to parse, summarise, synthesise, and survey academic PDF papers, with map-reduce chunking for papers exceeding the LLM context window
- Orchestrated dual Ollama instances on separate ports – a lightweight Llama 3.2 for routing and Llama 3.1 for task execution – to decouple decision-making from work and prevent LLM blocking
- Implemented full observability by logging every agent decision, LLM prompt, and output to timestamped JSONL trace files per run
- Generated structured mini-surveys (≤800 words) with inline citations synthesised across multiple research papers

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Citestat – Academic Citation Analytics Dashboard *Web Development*

Keywords: React, TypeScript, Vite, Tailwind CSS, Chart.js, Reagraph, Crossref API, OpenCitations API **Live demo** **Project repository**

- As part of a 3-person team, built a client-side citation analytics dashboard that queries Crossref and OpenCitations REST APIs for paper and author metadata
- Visualised yearly citation trends as histograms using Chart.js and citation network graphs using Reagraph (backed by D3.js)
- Computed bibliometric impact metrics (h-index, m-quotient) fully in the browser with no backend dependency

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Real-Time Collaborative Text Editor using CRDT *Distributed Systems*

Keywords: TypeScript, React, CodeMirror 6, WebSocket, CRDT, RGA, LWW Register, IndexedDB **Project repository**

- As part of a 3-person team, implemented an operation-based RGA CRDT for document text, ensuring convergence across concurrent edits via vector clocks and causal delivery
- Implemented a state-based LWW Register CRDT for metadata (title, cursors), with timestamp + siteId total order for deterministic conflict resolution
- Built a WebSocket relay server and React/CodeMirror 6 browser editor with IndexedDB-backed offline editing and automatic reconnect sync
- Enforced per-message access control using HMAC-signed role tokens, applied at the relay so role changes take effect immediately without reconnection

Distributed k-NN & Task Queue over gRPC

Distributed Systems, Concurrency

Keywords: Python, gRPC, Protocol Buffers, Distributed Systems, Systems Programming, Concurrency **Project repository**

- Built two distributed systems over gRPC/Protocol Buffers: a master–worker k-NN engine and a fault-tolerant task-queue manager
- Partitioned datasets across worker servers, broadcast queries in parallel, and merged per-worker results into the global top-k by Euclidean distance
- Pushed CPU/IO tasks to capability-matched workers from FIFO queues using server-side streaming RPCs
- Implemented heartbeat-based failure detection that re-queues in-progress tasks from dead workers onto healthy ones

Parallel Minimum Spanning Tree (MPI + SLURM)

Parallel Computing, Algorithms

Keywords: C++, MPI, SLURM, Parallel Computing, Union-Find **Project repository**

- Implemented parallel Boruvka's MST, distributing graph edges across MPI processes on a SLURM cluster
- Drove each round with MPI collectives (Scatterv, Allreduce, Allgatherv) for minimum-edge selection and component merging
- Tracked connected components using Union-Find with path compression and union by rank
- Built a performance harness reporting speedup, parallel efficiency, and computation-vs-communication profiling

LLVM Optimization Passes

Compilers

Keywords: C++, LLVM, Compiler Design, Static Analysis, Dataflow Analysis **Project repository**

- As part of a 2-person team, built out-of-tree LLVM passes on the new pass manager, compiled as shared-library plugins loaded via `opt`
- Implemented analysis passes: control-flow-graph DOT printing and static per-function call counting
- Implemented transformation passes over LLVM IR: constant folding/propagation and identity-operand instruction rewriting
- Implemented dead-code elimination using a fixed-point live-variable dataflow analysis

Racket-to-LLVM Compiler Frontend

Compilers, Type Systems

Keywords: Programming Languages, C++, LLVM, Compiler Design, Recursive-Descent Parsing, Code Generation **Project repository**

- As part of a 2-person team, built a compiler frontend translating a subset of Racket to LLVM IR (lexer, recursive-descent parser, semantic analyzer, code generator)
- Applied the visitor pattern for both static type checking and IR generation over the AST
- Resolved lexical scoping with a scope-stack symbol table and performed bottom-up static type checking
- Generated SSA-style IR with PHI nodes for conditionals, basic blocks for loops, heap-allocated tuples, and recursive functions

Bernstein Trajectory Fitting & CEM Obstacle Avoidance

Motion Planning, Optimization

Keywords: Robotics, Python, NumPy, Trajectory Optimization, Cross-Entropy Method **Project repository**

- Fit order-5 Bernstein polynomials to non-holonomic robot trajectories under velocity, acceleration, and orientation boundary constraints
- Solved per-axis constrained least-squares via the Moore–Penrose pseudo-inverse across under-, exactly-, and over-determined regimes
- Applied the Cross-Entropy Method (CEM) to sample waypoints yielding collision-free paths around static obstacles
- Validated collision avoidance on rolled-out trajectories and rendered simulation videos

Franka Robot Arm Motion Planning

Motion Planning

Keywords: Robotics, Python, PyBullet, Motion Planning, RRT-Connect, Sampling-based Planning **Project repository**

- Planned collision-free motion for a 7-DOF Franka Panda arm in a cluttered scene using bidirectional RRT (RRT-Connect)
- Implemented custom sample and extend functions in configuration space respecting URDF joint limits
- Performed edge collision checking and path interpolation/smoothing before execution
- Simulated and auto-recorded trajectory execution in PyBullet

Tabular Reinforcement Learning – Windy Gridworld

Reinforcement Learning, Algorithms

Keywords: Python, NumPy, Reinforcement Learning, Dynamic Programming, Q-learning **Project repository**

- As part of a 5-person team, implemented five tabular RL methods from scratch in NumPy (Value/Policy Iteration, Monte Carlo, SARSA(λ), Q-learning) for a stochastic Windy Gridworld
- Compared model-based planning against model-free learning on a 10×7 grid with stochastic upward wind

- Modeled the task as an infinite-horizon discounted MDP with an absorbing goal
- Analyzed convergence and instantaneous/cumulative regret against an exact Value-Iteration baseline

Certifications

- 🔗Working with containers: Docker, Docker compose, Docker Swarm (EDUCATIVE.IO)
- 🔗Flask Develop web applications in Python: Flask, Python, CSS, HTML (EDUCATIVE.IO)
- 🔗Java basic certification (HACKERRANK)

Skills

- **Programming and Databases :** Python, C/C++, Rust, Java, TypeScript, Dart, SQL, PostgreSQL, MongoDB
- **Web Technologies:** Flask, React, Node.js, Express, Flutter, HTML5, CSS, Javascript
- **DevOps & Tools:** BASH, Linux, Docker, Git, LLVM, MPI
- **Core Concepts:** Data Structures and Algorithms, Object Oriented Programming
- **AI Tools:** Claude, Perplexity

Accomplishments

- Accomplished 77 All India Rank in IIIT PGEE CSE 2024 exam
- Achieved 95.4 percentile in GATE exam CS/IT 2024
- Qualified Stage 1 of N.T.S.E. in Madhya Pradesh
- Zonal Gold medalist in SOF National cyber Olympiad Madhya Pradesh zone class 9th

Interests and Extracurricular Activities

- Playing badminton
- Participated in Discernment International Summit M.U.N. 2016-17 at Noida